Partial Fractions

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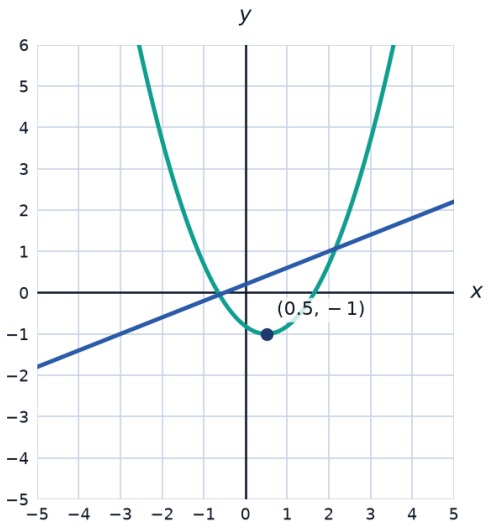
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Distinct linear factors

Express $(5x + 7)/((x + 1)(x + 2))$ in partial fractions.

2 [Foundational] • Target: Repeated linear factors

Express $(4x + 1)/(x - 1)^2$ in partial fractions of the form $A/(x - 1) + B/(x - 1)^2$.



3 [Foundational] • Target: Applying partial fractions to integration

Use partial fractions to evaluate $\int (5x + 7)/((x + 1)(x + 2))\,dx$, using the decomposition found earlier in this worksheet.

4 [Foundational] • Target: Applying partial fractions to a binomial series expansion

Express $1/((1 - x)(1 - 2x))$ in partial fractions, then use the result to write the first three terms of its series expansion in ascending powers of x.

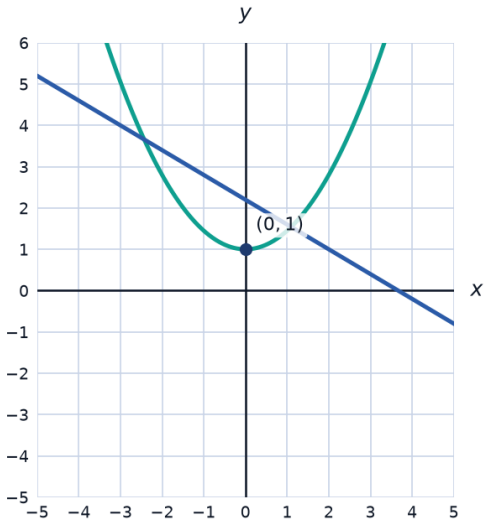
5 [Foundational] • Target: Combining terms before decomposing

Express $(x^2 + x + 1)/(x(x + 1))$ in partial fractions. (Hint: the degree of the numerator equals the degree of the denominator, so first perform polynomial division.)

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6 [Foundational] • Target: Applying partial fractions to a rate problem

A population model gives the rate of change $dP/dt = 1/(P(10 - P))$. Express $1/(P(10 - P))$ in partial fractions of the form $A/P + B/(10 - P)$.



7 [Foundational] • Target: Partial fractions with an improper fraction and repeated factor

Express $x^3/(x - 1)^2$ in partial fractions, first performing polynomial division since the numerator has higher degree than the denominator.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using partial fractions to sum a series

Express $1/(n(n + 1))$ in partial fractions, and use the result to find $1/(1 \cdot 2) + 1/(2 \cdot 3) + 1/(3 \cdot 4) + 1/(4 \cdot 5)$ by telescoping.

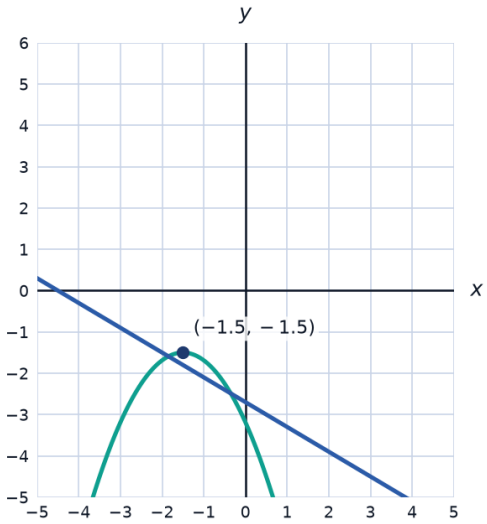
9 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

Use partial fractions to find the Maclaurin (power series) expansion of $1/((1 - x)(2 - x))$ up to the term in x^2, first decomposing into the form $A/(1 - x) + B/(2 - x)$.

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10 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

Express $(4x^2 - 5x - 15)/((x + 1)(x - 2)(x + 3))$ in partial fractions, using the cover-up method (evaluating the numerator at each root while covering the matching factor in the denominator).



11 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

A chemical reaction's rate is modeled by $dC/dt = 1/((C + 2)(C - 1))$, where C is a concentration. Express $1/((C + 2)(C - 1))$ in partial fractions of the form $A/(C + 2) + B/(C - 1)$, and then use the result to write the general antiderivative $\int 1/((C + 2)(C - 1))\,dC$ in terms of natural logarithms.

12 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

Use the decomposition $1/(3x) + 8/(3(x - 3))$ found earlier in this worksheet for $(3x - 1)/(x(x - 3))$ to evaluate $\int_4^5 (3x - 1)/(x(x - 3))\,dx$, giving your answer as a single logarithm expression.

13 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

This question has three parts. (a) Factor $x^2 - 4$ and hence express $(x + 8)/(x^2 - 4)$ in partial fractions. (b) Use your result from part (a) to write down $\int (x + 8)/(x^2 - 4)\,dx$. (c) Hence evaluate $\int_3^5 (x + 8)/(x^2 - 4)\,dx$, giving your answer as a single expression involving natural logarithms.

14 [Intermediate] • Target: Combining partial fractions back into a single fraction

Combine $2/(x + 1) + 3/(x + 2)$ into a single fraction.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Solving a separable differential equation using partial fractions • Cross-Lesson: 1.01 Proof by Mathematical Induction

Solve the differential equation $dP/dt = P(10 - P)/100$, given that $P = 5$ when $t = 0$. Use the partial fraction decomposition $1/(P(10 - P)) = 1/(10P) + 1/(10(10 - P))$ found earlier in this worksheet to integrate, and give P explicitly as a function of t.

16 [Challenging] • Target: Explaining when polynomial division is required • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Explain why polynomial division must be performed before applying partial fraction decomposition to $x^3/(x - 1)^2$, but not to $(3x + 1)/(x - 1)^2$.

17 [Challenging] • Target: Verifying a decomposition by substitution • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Verify the decomposition $(5x + 7)/((x + 1)(x + 2)) = 2/(x + 1) + 3/(x + 2)$ found earlier in this worksheet by substituting $x = 0$ into both sides.

18 [Challenging] • Target: Choosing efficient substitution values • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

When decomposing $(2x + 3)/((x - 4)(x + 5))$ using the equation $2x + 3 = A(x + 5) + B(x - 4)$, which two values of x should be substituted to find A and B most efficiently, and why?

19 [Challenging] • Target: Distinct linear factors • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.

20 [Challenging] • Target: Repeated linear factors • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying the correct form of a decomposition

Which of the following is the correct form for the partial fraction decomposition of $(3x + 1)/((x - 2)^2(x + 1))$?

a. A/(x-2)+B/(x-2)^2+C/(x+1)

b. A/(x-2)+B/(x+1)

c. A/((x-2)^2)+B/(x+1)

d. (Ax+B)/(x-2)^2+C/(x+1)

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22 [Foundational] • Target: Repeated linear factors

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

a. -3

b. 1

c. 3

d. 24

23 [Foundational] • Target: Irreducible quadratic factors

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

24 [Foundational] • Target: Applying partial fractions to integration

The rational function f is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x-2)(x^2+1)}$, where k is a real constant. Given that the series expansion of $f(x)$ in ascending powers of x, valid for sufficiently small $|x|$, has no term in x^2, find the value of k.

a. $\frac{25}{4}$

b. $\frac{17}{4}$

c. $-\frac{25}{4}$

d. $\frac{115}{12}$

25 [Foundational] • Target: Applying partial fractions to a binomial series expansion

The first three terms of the binomial expansion of $(1 + kx)^n$ are $1 + 6x + 12x^2$. Find the value of k.

a. 2

b. -2

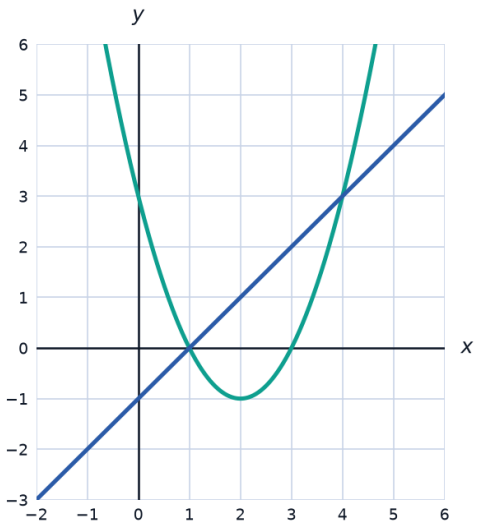
c. 3

d. -3

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26 [Foundational] • Target: Combining terms before decomposing

Consider the quadratic function $f(x) = x^2 - 4x + 3$ and the linear function $g(x) = x - 1$. The points of intersection are given by their solutions. What is the y-coordinate of the point of intersection where x = 1?



a. 0

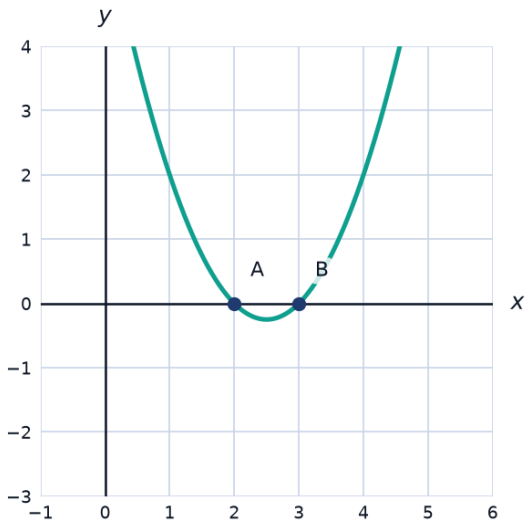
b. 1

c. -1

d. 2

27 [Foundational] • Target: Applying partial fractions to a rate problem

The graph of the quadratic function $f(x) = x^2 - 5x + 6$ is shown below. The graph intersects the x-axis at points A and B. Find the sum of the x-coordinates of points A and B.



a. 5

b. -5

c. 6

d. -6

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28 [Foundational] • Target: Partial fractions with three distinct linear factors

Generalised binomial: $\sqrt{1-x} = \sum_{n=0}^{\infty} \binom{1/2}{n}(-1)^n x^n$. First three terms:

a. $1 - x - x^2$

b. $1 + x/2 - x^2/8$

c. $1 - x/2 - x^2/8$

d. $1 - x/2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Partial fractions with an improper fraction and repeated factor

Find $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ using partial fractions:

a. 1

b. 1/2

c. 2/3

d. 3/4

30 [Intermediate] • Target: Using partial fractions to sum a series

Partial fractions: if denominator has irreducible quadratic $x^2 + bx + c$, the numerator is:

a. A

b. $A/(x^2 + bx + c)$

c. $Ax^2 + B$

d. $Ax + B$

31 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

Quadratic factor in partial fractions: $\frac{1}{x^2+4}$ gives integral:

a. $\frac{1}{2}\arctan(x/2) + C$

b. $\arctan x + C$

c. $\ln(x^2 + 4) + C$

d. $1/(2x + 4) + C$

32 [Intermediate] • Target: Combining partial fractions back into a single fraction

The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = ax^2 + bx + c$. Given that the vertex of the parabola is at $(-1, 8)$ and the graph passes through the point $(1, 0)$, find the values of a and b.

a. a = -2, b = -4

b. a = -2, b = 4

c. a = 2, b = 4

d. a = 2, b = -4

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33 [Intermediate] • Target: Identifying the correct form of a decomposition

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, 1, -4)$, $B(1, 4, -2)$, and $C(-1, -3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 6

b. 9

c. 7

d. 14

34 [Intermediate] • Target: Solving a separable differential equation using partial fractions

A sphere's radius increases at $\frac{dr}{dt} = 3$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 5$ cm.

a. 320π cm^3/s

b. 310π cm^3/s

c. 300π cm^3/s

d. 280π cm^3/s

35 [Intermediate] • Target: Explaining when polynomial division is required

A line has vector equation $\mathbf{r} = (-1; 4; 6) + t(-1; 4; 2)$. Find the point when $t = -4$.

a. $(3, -12, -2)$

b. $(4, -10, -3)$

c. $(2, -10, 0)$

d. $(5, -14, 0)$

36 [Intermediate] • Target: Repeated irreducible quadratic factor

An investment grows by 3%, 2%, 18% in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 8.93\%

b. 5.93\%

c. 9.43\%

d. 7.43\%

37 [Intermediate] • Target: Verifying a decomposition by substitution

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Choosing efficient substitution values • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Which of the following is the correct expression of $\frac{5x+1}{(x-1)(x+2)}$ in partial fractions?

a. $\frac{2}{x-1}+\frac{3}{x+2}$

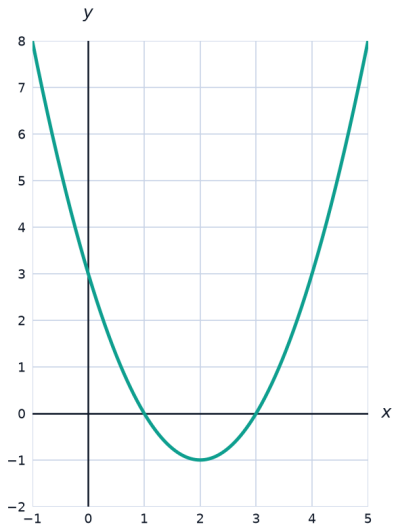
b. $\frac{3}{x-1}+\frac{2}{x+2}$

c. $\frac{2}{x-1}-\frac{3}{x+2}$

d. $\frac{1}{x-1}+\frac{4}{x+2}$

39 [Challenging] • Target: Distinct linear factors • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = x^2 - 4x + 3$. Identify the coordinates of the local minimum of the graph.



a. (2, -1)

b. (1, 0)

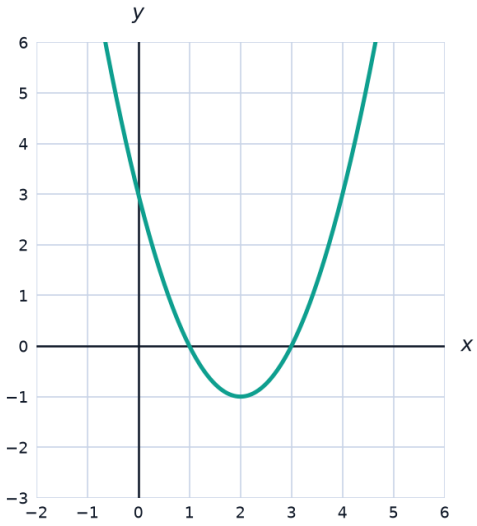
c. (3, 0)

d. (2, 1)

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40 [Challenging] • Target: Repeated linear factors • Cross-Lesson: 3.01 3D Vectors & the Cross Product

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the x-coordinate of the vertex of the parabola.



a. 1

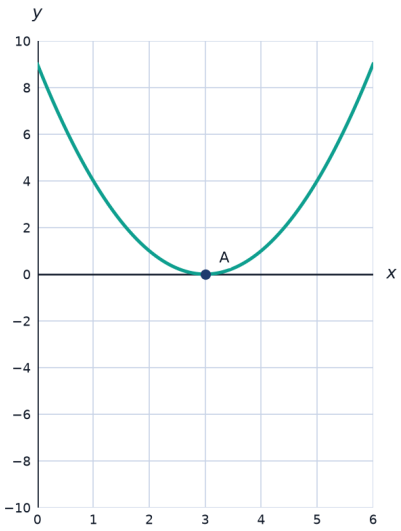
b. 2

c. 3

d. 4

41 [Challenging] • Target: Irreducible quadratic factors • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

The diagram shows the graph of a quadratic function $y = f(x) = x^2 - 6x + k$. The graph intersects the x-axis at point A, and has a minimum point at B. The coordinates of point A are $(3, 0)$. Find the value of k.



a. -9

b. 0

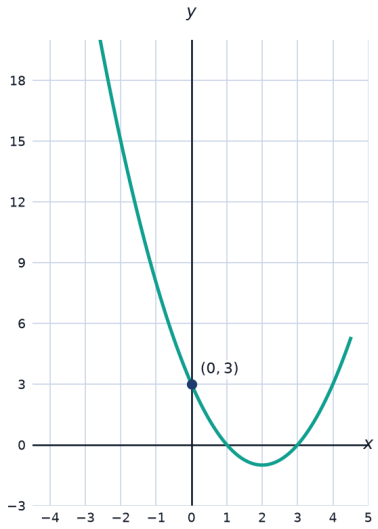
c. 9

d. 18

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42 **[Challenging]** • **Target: Applying partial fractions to integration** • **Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

Consider the graph of the quadratic function $y = x^2 - 4x + 3$. What are the coordinates of the vertex of this parabola?



a. (2, -1)

b. (1, 0)

c. (3, 0)

d. (2, 1)

43 **[Challenging]** • **Target: Applying partial fractions to a binomial series expansion** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

$\int_0^{\infty} \frac{1}{1+x^4}\,dx$: evaluate using partial fractions over $\mathbb{R}$:

a. $\pi/2$

b. $\pi/4$

c. $\pi/(2\sqrt{2})$

d. π

44 **[Challenging]** • **Target: Combining terms before decomposing** • **Cross-Lesson: 4.02 The Poisson Distribution**

Approximate $\int_0^1 \frac{\sin x}{x}dx$ using power series (first three terms):

a. $1 - 1/3 + 1/10$

b. $1 - 1/18 + 1/600$

c. $1 - 1/6 + 1/20$

d. $1 - 1/18$

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45 [Challenging] • Target: Applying partial fractions to a rate problem • Cross-Lesson: 5.01 Further Integration Techniques

Find the constant term of the quotient when $4x^2 + 6x + 3$ is divided by $(x + 1)$.

a. 2

b. 1

c. 4

d. 3

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